UNIT – IV

1. What is aerodynamics? Explain its significance in engineering applications.
2. Why is the study of aerodynamics important in the field of mechanical and aerospace engineering?
3. Differentiate between internal and external aerodynamics with suitable examples.
4. Describe the basic geometry of an aerofoil and explain its significance in aerodynamics.
5. Explain the basic working principle of a turbojet engine.
6. What are the key components of a gas turbine engine? Explain their functions.
7. What is an electric vehicle (EV)? Explain its significance in modern transportation.
8. Describe the basic architecture of an electric vehicle with a block diagram.
9. What are the key differences between conventional internal combustion engine (ICE) vehicles and electric vehicles?
10. What are the essential components of an electric vehicle? Explain their functions.
11. Discuss the factors affecting the performance of electric vehicles.
12. What are the different types of electric vehicle powertrains, and how do they affect performance?
13. What is a traction motor? List of different types of traction motors used in EVs.
14. Define tractive effort and explain its significance in vehicle motion.
15. What factors influence the tractive effort required by an electric vehicle?
16. How does regenerative braking contribute to energy efficiency in EVs?
17. How does aerodynamics impact the performance and efficiency of an EV?
18. What are the strategies used to improve the energy efficiency of electric vehicles?
19. Discuss the environmental benefits of electric vehicles over conventional fuel-based vehicles.
20. What are the major limitations and challenges faced in the widespread adoption of electric vehicles?